

FTRANSFER.CLOCKS.01 — the frozen external-clock contract

for the F_{transfer} jets

Preregistration companion (design, type [X]); frozen 2026-08-04

Status. Experiments-only design document (`experiments/ftransfer-clocks/`); the frozen preregistration of record is `hypotheses/ftransfer_clocks_v1.yaml`. Nothing here is load-bearing; no ledger row, no marker moves, no paper citation until a future promotion round carries the executed contract into `tftpt_research_contracts.tex`.

1. Where the contract starts

The F_{transfer} interface (`CONTRACT.F.01`) has its PGL_2 *syntax* closed: one common parabolic action on the discrete half, and on the native continuous half the classified degeneration $g_{\text{Boltzmann}} = g_{\text{pole}}$ (hyperbolic, forced identical by the single-flow theorem `v425`), g_{QCD} parabolic and exactly conjugate to the base translation, g_{relic} the identity (`v632.ftransfer_pgl2.py`). At jet level the native wall-time Schwarzian quadruple is

$$\left\{ -\frac{\Delta^2}{2}, -\frac{\Delta^2}{2}, 0, \text{degenerate} \right\}, \quad \Delta = 6 \ln \frac{3}{2}, \quad (1)$$

each flow Möbius *in its own clock* (`v578.native_jets.py`): the obstruction to a shared projective dynamics is the *clock*, not the projective structure. Both internal-clock rescues are machine-killed: strong thermal time (a modular generator is diagonalizable; the QCD clock map is parabolic with no real-diagonalizable logarithm, `v723`), and conditioned modular flows of one internal mother state (loxodromic, wrong classes, `v724`). *Diagnosis*: the clocks must be external — and without clocks fixed *in advance*, every jet statement is arbitrarily reparametrizable. This contract fixes the clocks.

2. The four frozen clocks (one per solver, no residual freedom)

Solver	Clock	Operational definition	Chart / gauge (frozen)
F_{pole} (Koide)	proper time τ	rest-frame proper time of the lepton one-particle state = the phase clock of the propagator pole, $e^{-im_\ell\tau}$ (on-shell scheme; PDG pole masses)	$x_1 = \ln(m_{\text{ref}}\tau)$; one <code>v99</code> seam tick = one proper-phase e-fold
$F_{\text{Boltzmann}}$ (η_B)	inverse temperature β	$\beta = 1/T$, FRW radiation plasma; adiabatic chart $g_* T^3 a^3 = \text{const}$ anchored at FIRAS $T_0 = 2.7255(6)$ K	$x_2 = \ln(M_1\beta)$; seam contraction $(2/3)^6$ per β e-fold
F_{QCD} (m_p/m_e)	RG time t	$t = \ln(\mu/\mu_0)$, $\overline{\text{MS}}$, $\mu_0 = M_Z$ frozen; 1-loop object $\dot{\alpha}_s = -\frac{b_0}{2\pi}\alpha_s^2$, $b_0 = 7$; window $\mu \in [m_t, 1 \text{ TeV}]$	identity gauge
F_{relic} (axion)	e-folds $\ln a$	$a_0 = 1$; Planck 2018 background; $\theta = \theta_i$ pre-onset, $n_a a^3$ const post-onset ($3H(T_{\text{osc}}) = m_a$)	degenerate channel: scored on kills only

The Koide clock is made precise deliberately: the on-shell (pole) chart *is* the proper-time chart, and this assignment is falsifiable — $Q = \frac{2}{3}$ holds for pole masses ($Q_{\text{exp}} = 0.6666645(51)$, pull -0.43σ) and fails by a factor $\gtrsim 500$ for running $\overline{\text{MS}}$ masses at M_Z ; the preregistration requires $|Q_{\text{run}}(M_Z) - \frac{2}{3}| > 10|Q_{\text{pole}} - \frac{2}{3}|$ as a dictionary check. The excluded alternative gauge for $F_{\text{Boltzmann}}$ (contraction per unit $z = M_1\beta$ instead of per β e-fold) is named in the YAML so it cannot be adopted post hoc.

3. Transport and the preregistered prediction

The common projective connection lives on the log-scale line $\sigma = \ln(\mu/M_Z)$. The four dictionaries are affine in their log charts with slopes $(-1, -1, +1, -1)$ *frozen*; every dimensionful anchor (m_{ref} , M_1 , μ_0 , a_0 , the thermal matching constant $\xi = 1$) enters as an affine offset only and drops out of every Schwarzian and every conjugacy class — the contract tests exactly the anchor-free jet content, consistent with the `v723` anchor census $\{\Lambda_{\text{QCD}} \text{ unit}, \theta_i, C_p\}$ staying external. Each channel exports the order-3 jet (y, y', y'', y''') of its native trajectory in its own frozen clock at the frozen base point; transport is the Schwarzian cocycle

$$\{y \circ h, \sigma\} = (\{y, x\} \circ h) (h')^2 + \{h, \sigma\}, \quad \{e^{a\sigma}, \sigma\} = -\frac{a^2}{2}, \quad (2)$$

with h the frozen dictionary. All kernel steps are symbolic (sympy exact); data enter only through base points and dictionary-validity checks.

Prediction FTC.P1 (frozen). There exists *one* constant-coefficient Riccati equation $y' = c_0 + c_1 y + c_2 y^2$ on σ satisfied by all non-degenerate transported jets. Since all non-constant solutions of one such equation share the constant Schwarzian $S = -(c_1^2 - 4c_0c_2)/2$, this is equivalent to: the transported Schwarzians of F_{pole} , $F_{\text{Boltzmann}}$, F_{QCD} are *equal* — the native quadruple (1) collapses into *one* PGL_2 class under the fixed physical clocks. The seam channels transport to $\{q, \sigma\} = \{y_B, \sigma\} = -\Delta^2/2$ exactly, so the prediction forces

$$\{\alpha_s, \sigma\} = -\frac{\Delta^2}{2} = -18 \ln^2 \frac{3}{2} = -2.9592351700 \dots \quad (3)$$

If it does not hold, the contract dies. The degenerate F_{relic} carries no class information (its content is the anchor θ_i); it is scored on the kill criteria and dictionary validity only — fixed-point membership of a constant is *not* scored, because with a per-channel Möbius frame it would be vacuous.

4. Kill criteria (frozen)

- K1** NO-COMMON-CONNECTION: with the clocks and dictionaries exactly as frozen, the transported Schwarzians differ (symbolic inequality, no tolerance). On kill, the executor must exhibit the reparametrization that *would* be needed per failing pair, documenting the forbidden freedom.
- K2** FREE-REPARAM-NEEDED: a pass is achievable only by a post-freeze, per-channel change of gauge or dictionary (re-sloped or non-affine dictionary, either e-fold gauge, ξ , μ_0 , scheme, window, frame re-fit). Machine form: the executor runs against the byte-frozen YAML (sha256 recorded); any deviation needed for a pass is K2.
- K3** SECOND-CLOCK-NEEDED: any channel needs a second time variable to express its exported jet (e.g. F_{QCD} needing both μ and T). Each export must be a function of exactly one clock, asserted structurally on the symbolic exports.

5. Decision rules and executor

Data (pin base points and dictionary validity only): PDG 2024 pole masses ($m_\tau = 1776.93(9)$ MeV dominates; frozen tolerance $|q_{\text{exp}} - 2| \leq 3\sigma_q$, $\sigma_q = 1.52 \times 10^{-5}$); FIRAS T_0 + Saikawa–Shirai g_{*s} table + Planck $N_{\text{eff}} = 2.99(17)$ for the β chart; $\alpha_s(M_Z) = 0.1180(9)$ (PDG world average) with the frozen 1-loop chain $\alpha_s^{(1\ell)}(m_t) = 0.10807$ and a 10% overlay tolerance on $[m_t, 1 \text{ TeV}]$ (honestly covering the ≥ 2 -loop truncation, where the flow is no longer Riccati at all — a named, frozen limitation); Planck 2018 background for $\ln a$ with slope tolerance $|\text{d}\sigma/\text{d}\ln a + 1| \leq 0.15$ across the onset window. A broken dictionary is CONSTRUCTION-FAIL, not a physics kill.

Controls: identity dictionaries must reproduce (1) exactly (calibration); a deliberate clock swap (F_{QCD} dictionary on the Koide channel) must *fail*; a frozen Möbius re-framing must change nothing (cocycle correctness).

Executor: a future verification module (name pattern `vN_ftransfer_clock_jets.py`), N assigned at the next promotion round (≥ 15 ; round 14 runs on the suite surfaces), entered exclusively via the promotion workflow; verdict enum frozen as $\{\text{ONE-CLASS}, \text{NO-COMMON-CONNECTION}, \text{FREE-REPARAM-NEEDED}, \text{SECOND-CLOCK-NEEDED}, \text{CONSTRUCTION-FAIL}\}$. No execution before promotion.

6. Honest priors

Stated before execution (the v723 precedent: preregistering the expected kill is house style). The expected verdict is **K1**: all four frozen dictionaries are affine on the log-scale chart, affine transport scales a Schwarzian by $(\text{slope})^2$ and cannot move 0 (the QCD value) onto $-\Delta^2/2$ (the seam value) — the v578 split survives every slope, and the no-diagonalizable-logarithm kill of v723 is the operator-side shadow of the same fact. The survival branch would require a non-affine *physical* dictionary for at least one channel, and none is admissible under the frozen operational definitions. Executing anyway is the point: a K1 kill makes the clock obstruction *final* at the physical-clock level — the jet route of **CONTRACT.F.01** closes, and the anchor census $\{\Lambda_{\text{QCD}} \text{ unit}, \theta_i, C_p\}$ stands as the full residual — while ONE-CLASS, the only surprising outcome, would falsify this expectation and revive a shared projective dynamics. Either way the answer is forced, which is what freezing the clocks is for.